UMC 110nm Logic and Mixed-Mode Al Advanced Enhancement 1.2V PCAP SPICE Model Document

Version 0.1 Phase2
2010/11/02

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
V0.1p1	2010/08/31	Shih-Hsin Yeh	1. Original release based on new DSM definition form 2009/11/30.
0.1_p2	2010/11/02	Shih-Hsin Yeh	1. Removed the description m in the model. 2. Revised the Igate formula in the model. 3. Modified the default nf value.

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 110 nm Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version 7.1_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

typ: Typical capacitor
max: Maximum capacitor
min: Minimum capacitor

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE
- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
.lib "./ModelFileName.lib" CaseName
, where CaseName could be typ, max, or min.
- Define device condition by 'Xxx' statement.
Example:
X1 VGG GND DeviceName Lf=2E-6 Wf=5E-6 NF=2

For Cadence Spectre
- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName could be typ, max, or min.
- Define device condition by 'Xxx' statement.
Example:
X1 (VGG GND) DeviceName Lf=2E-6 Wf=5E-6 NF=2

For Mentor Graphics Eldo
- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.
.lib "./ModelFileName.lib.eldo" CaseName
, where CaseName could be typ, max, or min.
- Define device condition by 'Xxx' statement.
Example:
X1 VGG GND DeviceName Lf=2E-6 Wf=5E-6 NF=2

ModelFileName: L110AE_PCAP12_V012
DeviceName: PCAP_12

Geometry description: (before 90% shrink)
0.15 µm <= Lf <= 2 µm
2.5 µm <= Wf <= 10 µm
2 <= Nf <= 8

Bias Range:
-1.2V <= VGS <= 1.2V
VDS = VBS = 0V

Temperature range:
25 °C

Layout example:

The following figure is the layout example of the PCAP. Lf and Wf are the single poly finger length and width, respectively, and Nf is the total finger number.

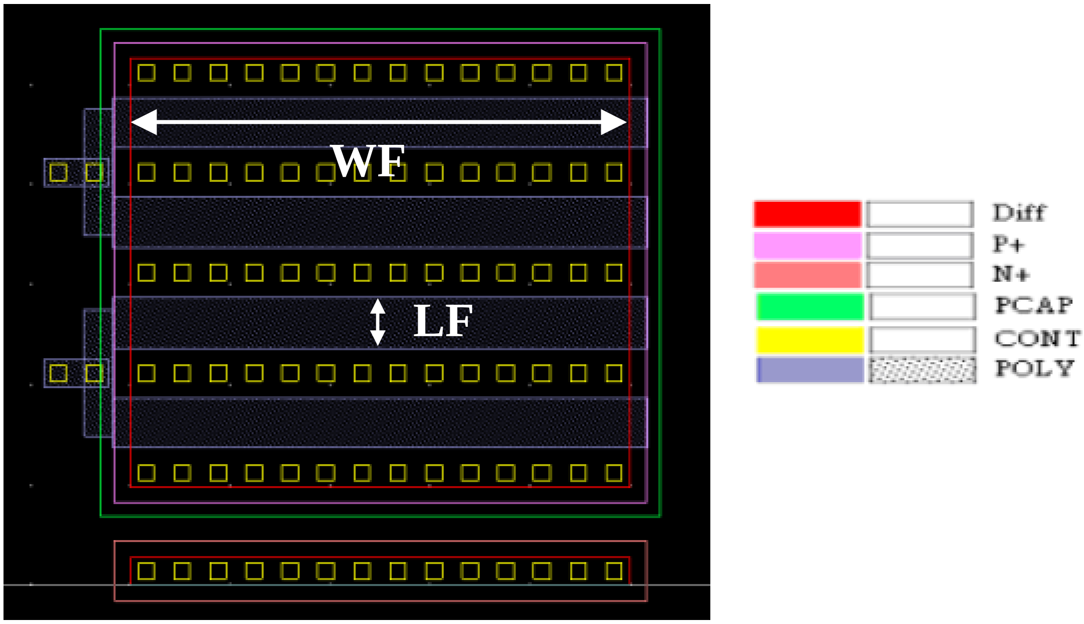


Figure 1-1 PCAP layout

2 Capacitance Model

2.1 Silicon Wafer for Modeling

The information of the hardware used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for PCAP

	Wafer for PCAPs measurements data
Mask ID	BSW0FA
Lot ID	D4YAA
Wafer Number	#7

2.1.1 Test devices

The following geometries were available for parameters extraction.

Table 2-2 Geometry of PCAP devices measured for parameters extraction.

W\L	2	1	0.5	0.25	0.15
10			X		
5	X	X	X	X	X
2.5			X		

2.1.2 Measurement conditions

The capacitance was measured under the conditions summarized in the table below.

Table 2-3 Test conditions for Cg curves of PCAP

	C_G vs. V_G
V_G	From 1.2 to –1.2V in –0.1V step

V_B	0V
Frequency [Hz]	500M Hz
Temp [°C]	25

2.2 Electrical Parameters

The following table shows the measured and simulated Cmax (fF), where Cmax is the capacitance at VG=-1.2V, VB=0 V, T=25°C. The dimension is after 90% shrink and 12nm sizing channel length back.

Table 2-4 Electrical Parameters for PCAP

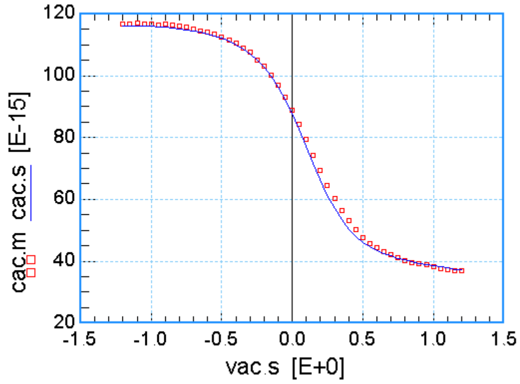
All parameters are given for T=25 ℃, unless specified otherwise.

Name	WF (um)	LF (um)	Nf	M	Measured Cmax (F)	Model Cmax (F)
PCAP_401	4.5	0.147	4	4	1.17e-13	1.16e-13
PCAP_403	4.5	0.237	4	4	1.87e-13	1.90e-13
PCAP_405	4.5	0.62	4	2	1.88e-13	1.88e-13
PCAP_407	4.5	0.912	4	1	1.87e-13	1.87e-13
PCAP_408	4.5	1.812	2	1	1.87e-13	1.85e-13
PCAP_409	2.25	0.462	4	4	1.91e-13	1.90e-13
PCAP_410	9	0.462	4	1	1.88e-13	1.88e-13
PCAP_411	4.5	0.462	2	4	1.89e-13	1.88e-13
PCAP_412	4.5	0.462	8	1	1.88e-13	1.88e-13

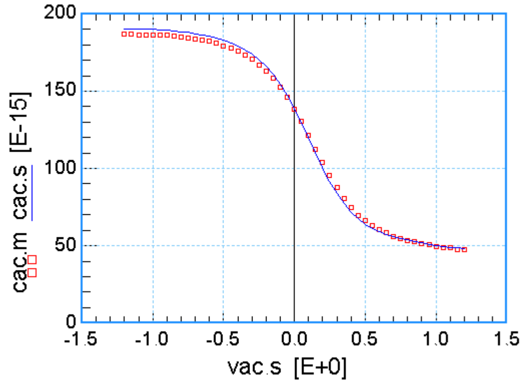
2.2.1 Cg vs. Vg plots

The following figures show the measured and simulated Cg (F) vs. Vg (V). The measurement data is dots and simulation is line, respectively.

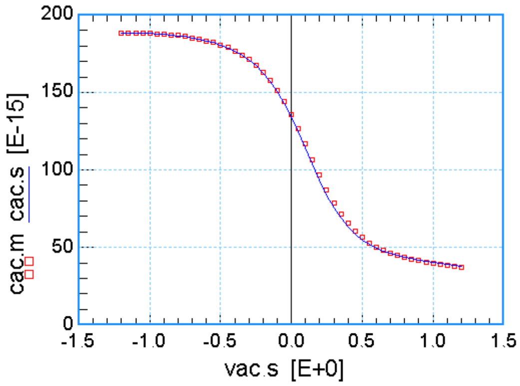
PCAP_401



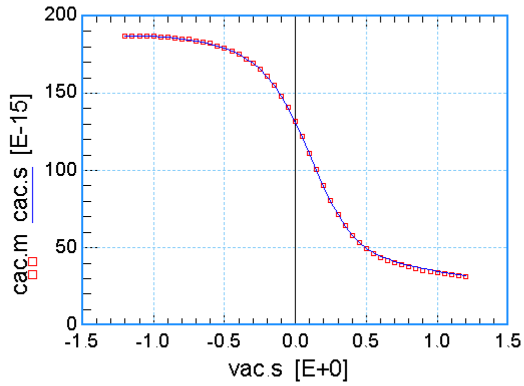
PCAP_403



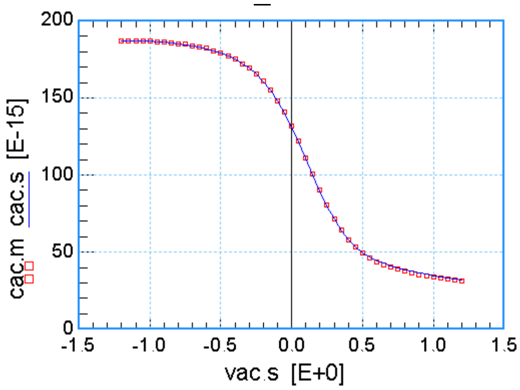
PCAP_405



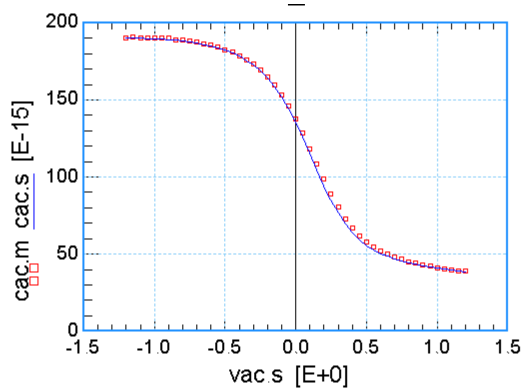
PCAP_407



PCAP_408



PCAP_409



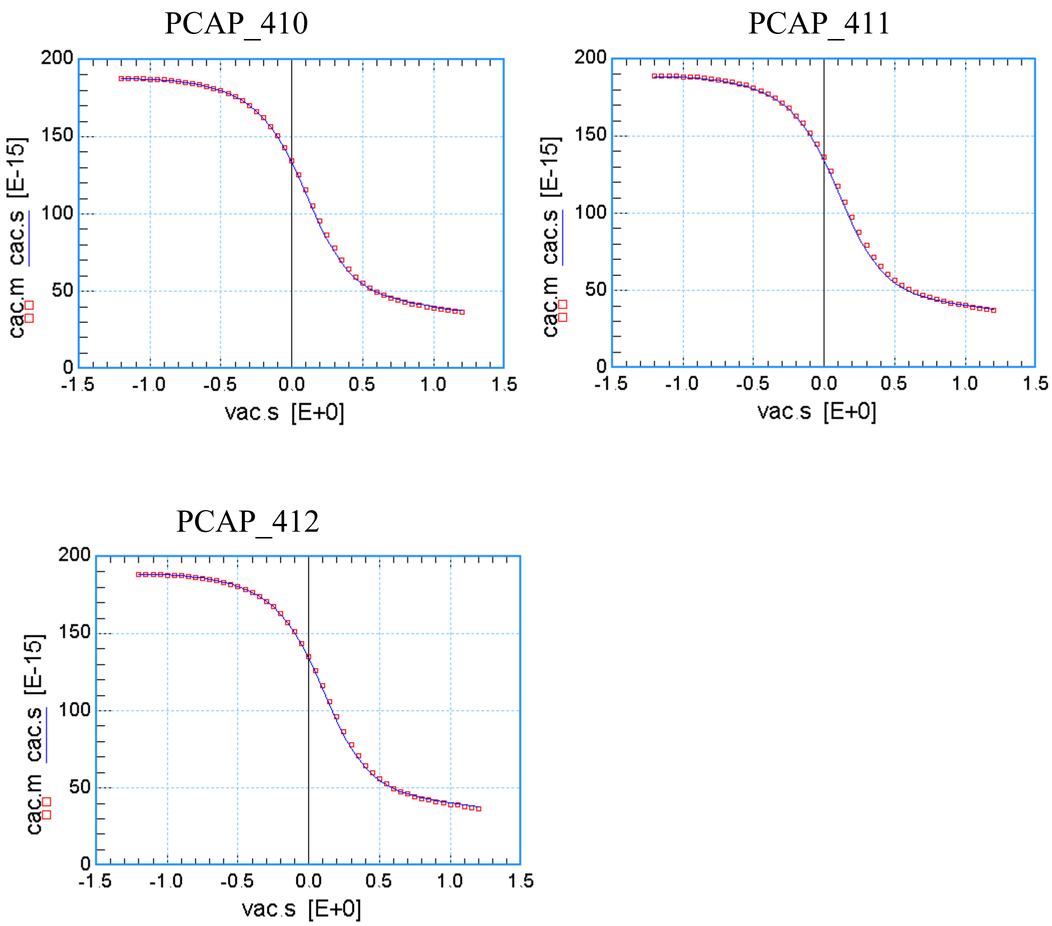
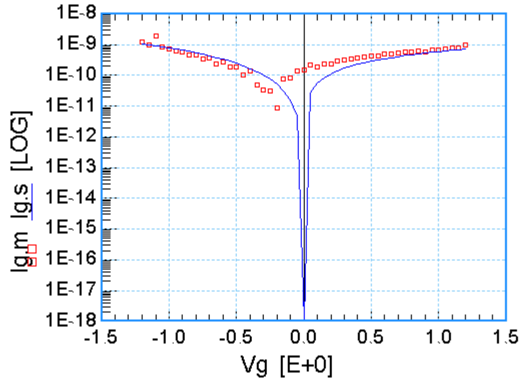


Figure 2-1 Cg vs. Vg at 25 ℃

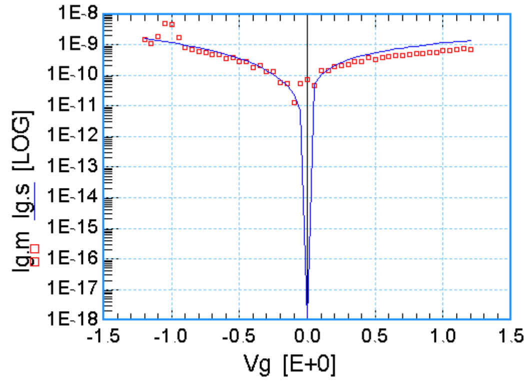
2.2.2 Igate vs. Vg plots

The following figures show the measured and simulated Igate (A) vs. Vg (V). The measurement data is dots and simulation is line, respectively.

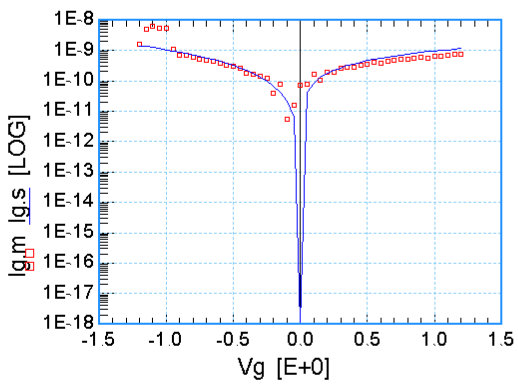
PCAP_401



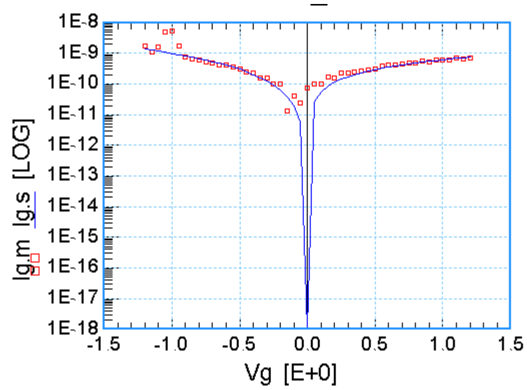
PCAP_403



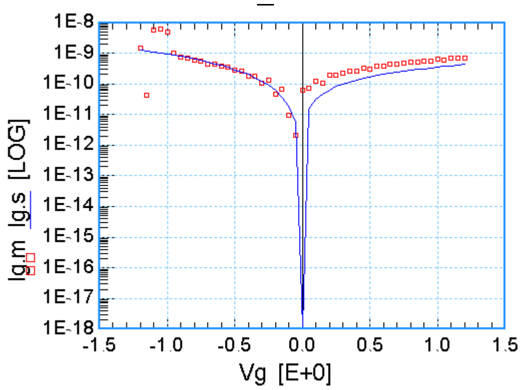
PCAP_405



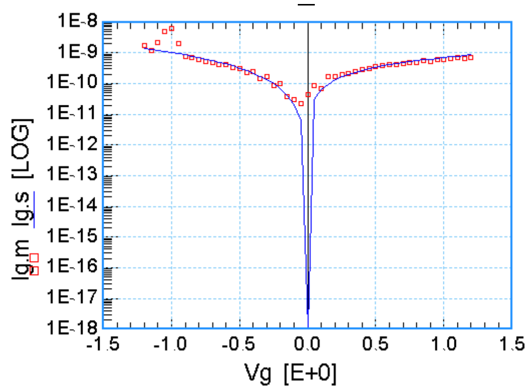
PCAP_407



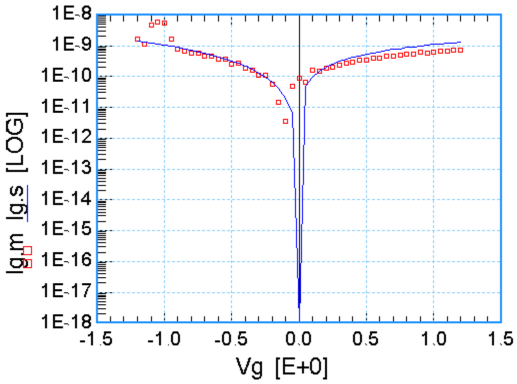
PCAP_408



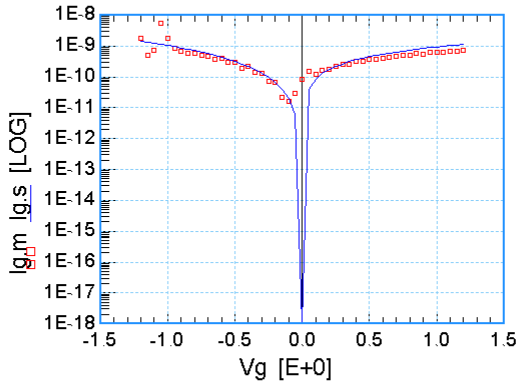
PCAP_409



PCAP_410



PCAP_411



PCAP_412

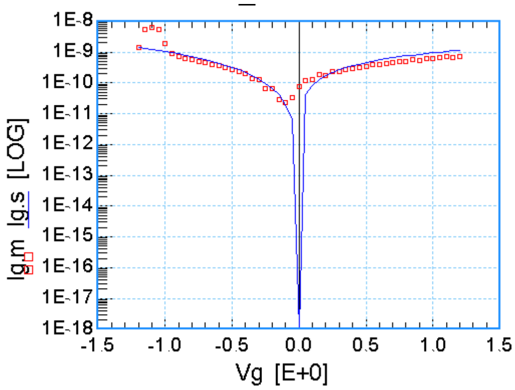


Figure 2-2 Igate vs. Vg at 25 ℃

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	None	None

3.2 ELDO Warning Message

Table 3-2 Eldo warning message

NO.	Warning Message	Explanation
1.	None	None

3.3 SPECTRE Warning Message

Table 3-3 Spectre warning message

NO.	Warning Message	Explanation
1	None	None